Digital optical microscope

How do digital optical microscopes work?

A digital microscope uses optics and a digital camera to capture images. Light hits the sample and travels through a series of lenses and is captured by a camera. This image is displayed on the computer.

The Breakerspace owns a DSX1000 Digital microscope. Capabilities include:

- Magnification from 14x to 5600x
- Can use a long working distance
- 3D imaging
- Automated capture and stitching of images in 2D or 3D mode
- Different lighting, including bright field, dark field, polarized, etc.
- Tiltable microscope head to view different angles
- Post-processing and analyzing tools!

What do you use each observation mode for?

Brightfield (BF)	For flat samples
Oblique	Similar to BF, emphasized scratched or unevenness
Darkfield (DF)	For finding edges, dust, or scratches, or for observing color that is difficult to observe in BF observation
Brightfield/darkfield mix	combines easy visibility in BF observation and high detectability in DF observation, recommended for searching scratches or defects that are difficult to find
Simple polarization	recommended for finding the best image of the sample, e.g., crystals of rocks or minerals, etc.
Differential interference contrast (DIC)	for observing fine unevenness (wave) that cannot be found with other observation methods

Instructions on using the Breakerspace Digital Microscope

Instrument startup:

- Turn machine on (the switch is at the back of the machine)
- Log on to instrument workstation using your MIT Kerberos
- Start the DSX Software and log on as Guest (no password)
- Acknowledge it is safe for the stage and the head to move
- Use the manual focusing knob to lower the microscope stage
- Load/change objectives if needed

Instrument operation:

- Put your sample on stage
  - Materials should be non-hazardous and safe to handle in the Breakerspace
  - Any liquids should be contained so they do not spill on the stage
  - Loading capacity of the stage is 5 kg
- Use the manual focusing knob to bring sample into focus
- With the joystick move stage back/forth and left/right to position desired sample area under observation
- Fine-tune focus by moving the zoom head with the console buttons/wheel or software buttons
- Select observation mode by clicking the best image button, select all modes, select desired image, and click apply
- Characterize as you wish!
- Make sure to save your files

Instrument shut-down:

- Close the DSX software
- Click yes to exit the microscope system and retract the head
- Once the software fully closes, switch the microscope off
- Log out of Windows
- Place the dust cover on the microscope

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!